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Proposal for the rules for the calculation of the Carbon Footprint of rechargeable Industrial Batteries except those with exclusively external storage (CFB-IND)

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Abstract

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1 Introduction

The new Batteries Regulation (Regulation (EU) 2023/1542) published the 28th of July 2023[1] sets rules on sustainability, performance, safety, collection, recycling, and second life of batteries as well as on information requirements about batteries. Among these, there is the carbon footprint of the batteries (CFB) to be determined and communicated as specified in Article 7 and the corresponding Annex II of the Regulation. The CFB quantifies the total amount of greenhouse gases as kg of CO2 equivalent per Functional Unit. The functional unit is defined as one kWh (kilowatt-hour) of the total energy provided by the battery system over the battery's service life, measured in kWh. For back-up batteries that have as their primary function to ensure continuity of a power source, the functional unit is defined as the ability to provide one kWmin (kilowatt-minute) of backup power capability at any moment over the lifetime of the battery.

The CFB shall be declared for "rechargeable industrial batteries with a capacity above 2 kWh, light means of transport (LMT) batteries and electric vehicle (EV) batteries placed on the Union market". CFB declarations will then be used to define the CFB performance classes and CFB thresholds that batteries entering the European market should comply with. Ultimately, this Regulation will ensure that the expected massive deployment of batteries (e.g., in mobility) will be associated with minimum overall carbon emissions. The CFB shall be accompanied by a public version and a non-public version of the CFB supporting study.

The Joint Research Centre (JRC) of the European Commission has been in charge of providing technical support to the development of the secondary legislation on the CFB (in line with the requirements of the Article 7 and Annex II of the Batteries Regulation). In June 2023, the JRC published the draft Rules for the CFB of EV batteries (CFB-EV),2 aiming at supporting the Delegated Act currently under development. The present document is a first working draft for setting the Rules for the CFB of industrial batteries with a capacity above 2 kWh, excluding those with exclusively external storage (CFB-IND), for which the CFB declaration becomes mandatory 12 month after that for EV batteries. The present report is based on the rules proposed for CFB-EV and it is a working document to be used as basis for on the discussion with stakeholders and policymakers.

Since many components of the CFB methodology are identical between all types of batteries in scope, the present working draft is elaborated only on aspects that are specific for industrial batteries and Table 1 summarises the major changes compared to the CFB-EV. In particular:

- The structure of this report is similar to the CFB-EV
- All the sections that are considered to remain 100% unchanged to the CFB-EV are left empty.
- In the other sections, all text that is expected to remain 100% unchanged compared to the former CFB-EV Report is written in **blue text** (and might be updated at later stage, based on the finalisation of the policy process for CFB-EV), while the modified text is in **black**.

NOTE: The text of carbon footprint rules has been kept as concise as possible. All sections specified as "NOTE" have to be considered as explicative to the text and not be intended as part of the rules for Carbon Footprint of Batteries (CFB).

NOTE: Specific questions that will be addressed at the stakeholder workshop on Dec. 12th 2023 are included in the document. These are highlighted by green text colour for easier identification.

Unless otherwise specified in this document, the EF method as in the EC Recommendation 2021/2279 shall be considered as general guidelines about how to determine the carbon footprint.

[1] EU, Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC, 2023. Available at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1542#:~:text=This%20Regulation%20imposes%20battery%20due,or%20putting%20them%20into%20service .

[2] JRC study in support of developing Carbon Footprint rules of batteries https://eplca.jrc.ec.europa.eu/EU_BatteryRegulation_Art7.html

47 **Table 1.** Major changes compared to Carbon Footprint Rules for EV Batteries (CFB-EV). CFF: Circular footprint formula, LIB:
48 Lithium-Ion Battery.

Section	**Major changes compared to CFB-EV**
2 Scope	Completely new.
3 Functional unit and reference flow 3.1 Functional unit for repetitive energy supply (REP) industrial batteries 3.2 Functional unit for on-demand (OND) or backup industrial batteries	Completely new, with functional unit definitions specific for industrial batteries.
4 System Boundaries and cut-off rules	Added one system component level for industrial batteries (container), including thermal management and fire protection system.
4.1 Cut-off rules	Added power conversion system / inverter and system installation to cut-off.
4.2 Raw material acquisition and pre-processing	Identical.
4.3 Manufacturing	Added system assembly.
4.4 Distribution	Some minor industrial batteries-specific aspects added (e.g., distribution via retail).
4.5 End-of-life	Practically identical.
5 Impact assessment	Identical.
6 Inventory data - Data collection requirements and quality requirements	
6.1 Mandatory company-specific processes	Identical.
6.2 Non-mandatory company-specific processes	Added specific materials for industrial batteries to the most relevant processes.
6.3 CFB-compliant company-specific datasets	Identical.
6.4 Requirements for the collection of company-specific data	Identical.
6.5 Data quality rating (DQR) of the CFB declaration and of CFB-compliant datasets	Identical.
7 Inventory data - Modelling requirements	
7.1 Electricity modelling	Identical.
7.2 Allocation rules	Virtual housing not applicable. Otherwise identical.
7.3 Recycling content and end-of-life (EoL) modelling	Additional CFF parameters for industrial batteries-specific materials. Added modelling guidelines for end-of-life for non-LIB batteries (end-of-life modelling for LIB and similar identical)
7.4 Transport	Identical
8 Verification	Identical
8.1 Application for verification	Identical
8.2 Technical documentation	Slightly different documentation requirements due to different functional units
8.3 Verification and validation techniques	Identical

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8.4	Data confidentiality	Identical

1.1 Terminology

This report uses the following terminology to indicate the requirements, as:

— The term “shall” is used to indicate what is required in order to calculate the CFB.

— The term “may” is used to indicate an option that is permissible. Whenever options are available, the CFB supporting study shall include adequate argumentation and evidence to justify the chosen option.

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2 Scope

This document provides rules on how to calculate the carbon footprint of rechargeable industrial batteries with a capacity above 2 kWh, excluding those with exclusively external storage (CFB-IND) within the scope of the Article 7 of the Battery Regulation. The CFB-IND shall be declared for any industrial battery that falls within the scope of Article 7 of the Batteries Regulation i.e., rechargeable industrial batteries with internal storage and a capacity greater than 2 kWh, independent of its cell chemistry and of the specific application. The scope of the CFB-IND excludes batteries that have been subject to preparation for re-use, preparation for repurposing, repurposing, or remanufacturing, and that had already been placed on the market or put into service before undergoing such operations (according to the Article 7 (point 5) of the Batteries Regulation).

NOTE: Although the rules defined in this document are generic and applicable to any industrial battery with internal storage, independent of its chemistry, the following battery chemistries have been analysed for the testing and development of the present CFB-IND Rules (especially related to section 6.2 and 7.3), being currently considered the most relevant ones:

Lead-Acid (flooded, valve-regulated / sealed, tubular and all other forms - PbA)

Lithium-Ion (LIB), including all current LIB chemistries and solid-state LIB

Sodium High Temperature (Na-S, Na-NiCl)

Sodium-Ion (SIB), including aqueous sodium and hybrid ion batteries (ASIB)

Nickel-Cadmium (NiCd), Nickel-Zinc (NiZn) and Nickel-Metal Hydride (NiMH)

Other emerging chemistries like Mg-S, Aluminium, etc. could potentially become relevant in future, especially once achieving market maturity. The same applies to emerging anode / cathode materials for any of the listed battery chemistries and to other battery technologies. If deemed necessary, the CFB Rules may be updated correspondingly.

Question to stakeholders:

- Is there any other battery chemistry of particular relevance that should be considered or looked at more in detail?
- What additional definitions could be relevant to add compared to former rules on CFB-EV?

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3 Functional unit and reference flow

Two fundamentally different services can be provided by rechargeable industrial batteries:

1. The **provision of electricity** that was previously charged to the battery. While different applications have very different use profiles, they all require the battery to regularly discharge its energy content, leading to a more or less frequent cycling behaviour. Systems providing this type of service are considered to provide **repetitive energy supply (REP).** This can equally be energy supply **in mobile equipment (REP-MOB) or in stationary equipment (REP-STA)**

2. The **provision of back-up** or on-demand services, maintaining the battery in charged state and discharging only sporadically in case of power failure or blackout. In the extreme case, the grid failure or power outage never happens, and the battery is never cycled but just kept as safety reserve. Systems providing this type of service are referred to as **on-demand (OND) batteries**. Similarly to REP batteries, the OND service can be provided in **mobile (OND-MOB)** or in **stationary equipment (OND-STA)**.

NOTE: The classification is based on the IEC 63369 under development.

3.1 Functional unit for repetitive energy supply (REP) industrial batteries

For energy-providing (REP) batteries, the **functional unit (FU)** is defined (according to Annex II of the Batteries Regulation) as *one kWh (kilowatt-hour) of the **total energy** delivered by the battery over its service life, measured in kWh".*

The **total energy** (in kWh), equivalent to the **quantity of functional unit** (QUa) provided over **service life** is the total amount of electricity (in kWh) provided by the battery until reaching its End of Service Life (EoSL). It corresponds to the energy throughput according to Article 14 / Annex VII of the Battery Regulation at the end of life). The total energy shall be directly measured during the corresponding test procedure by integrating over the discharged energy or be calculated by multiplying (a) the service life in cycles with (b) the average amount of delivered energy over each cycle, considering the decreasing energy capacity over time.

There are different options for determining the service life. All options require determining the service life in cycles and in years. Article 10 and Annex IV of the Batteries Regulation require these to be reported. However, at present there is no harmonised standard for determining these. The deadline for CEN-CENELEC to draw up such (a) standard(s) for the purpose of Article 10 is 7 December 2025, well after the expected deadline of 18 February 2025 for the Commission to adopt the Delegated Act on the methodology for the calculation and verification of the carbon footprint of industrial batteries. Thus, it is likely that no harmonised standards will be available in time that can be used for the carbon footprint methodology. Therefore, the following options have been identified

Question to stakeholders:

- Would you agree with allowing two different approaches for measuring the total energy (i) direct measuring and integrating over the cycle life test, ii) calculation from number of cycles and average amount of energy provided per cycles)? If not, which approach would be more appropriate?

Option 'manufacturer declares'

The **service life** of a battery is defined as the Reference Service Life of the battery system for the application as described in the technical specification of the battery system as provided to the purchaser (i.e., the end-user). The corresponding use profile and reference service life specified for the battery or battery system within the specific application shall be provided in cycles and in years together with the corresponding total energy and the reference test used in the CFB supporting study. This option is built on the approach suggested by the IEC 63369 that is currently under development.

Option 'one standard for all'

The **service life** of a battery shall be determined according to IEC 62620 (6.6.1 – Endurance in cycles, applying the shortened test procedure) until reaching a remaining discharge capacity of 80% of the

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initial energy capacity (I.e., the rated energy capacity). If the EoSL criterium is not reached after 500 cycles, the endurance test shall be repeated. The total energy (and corresponding number of full discharge cycle equivalents) shall be determined by integrating over the product of discharge voltage and current over the whole cycle life test, or by multiplying the number of cycles with 90% of the energy discharged from the first cycle as linear interpolation. If after two endurance test procedures (i.e., after 1000 cycles) the EoSL criterium is not yet reached, linear extrapolation may be applied using the line with the highest negative slope between each of the following three data points: (i) initial capacity, (ii) remaining capacity after 500 cycles and (iii) remaining capacity after 1000 cycles.

Option 'application-specific standards'

The **service life** of a battery shall be determined by applying the test standards and EoSL criteria in Table 2, with application-specific standards given priority above battery-chemistry-specific standards (i.e., the most application-specific standard that is applicable to the given cell chemistry shall apply, and be prioritized over a more cell-chemistry specific, but less application-specific standard). For applications not covered by any of the specific standards contained in Table 1, the EoSL criteria and test procedure of IEC 62620 applies.

NOTE: Table 2 is work in progress based on ongoing review of existing standards for industrial batteries.

Table 2. Service life test procedure and EoSL criteria for different battery types and applications according to existing standards. For all test procedures, the total energy is defined as the total amount of energy provided by the battery until reaching the specified EoSL criterium. While applicable also to OND- batteries, the standard does not provide any calendar lifetime test, but only a test on the upper limit of storage temperature. It is therefore not applicable for determining OND battery service life

Application	**Standard**	**Service**	**Chem**	**EoSL criterium**	**Test**
PV off grid	IEC 61427-1-(2013)	REP-STA PVoffGrid	All	80% of initial capacity	(cycles) Service-specific load profiles sequence applied at 40 °C until reaching EoSL criterium
EES On grid (stationary)	IEC 61427-2-(2015)	REP-STA: Freq.Reg., LoadFllw., PeakShv, TimeShift	All	Voltage during discharge at rated power below specified limits of operating voltage	(cycles) Service-specific load profiles repeatedly applied at 25 °C until reaching EoSL criterium
Generic	IEC 62984-3:2020	REP-STA	Na-high temp	n/a. Energy capacity loss shall be less than 3/5% after 350 cycles	(cycles) 350 cycles at nominal discharge power to 20%SoC. Not repeated until EoSL
Backup / on-demand	IEC 60896-21/22:2004	OND-STA	PbA	80% of initial capacity at 1/3C discharge	(days) Store battery at float charge at 40 °C and determine days until EoSL
Generic industrial	IEC 62620 (2014)	REP-STA REP-MOB	LIB	60% of initial capacity	(cycles) 500 cycles at a fixed C-rate. *Not repeated until EoSL*

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Generic industrial	IEC 63115-1:2020	REP-STA REP-MOB	NiMH	66% of initial capacity	(cycles) 500 cycles at a fixed C-rate. *Not repeated until EoSL*
Aircraft	IEC 60952-1:2013 - Aircraft	OND-MOB REP-MOB	All, but focus-ed on PbA and NiCd	Voltage during test falls below minimum voltage (1.5V /cell for PbA and 0.8V/cell for NiCd	(cycles) 4x24shallow followed by one deep cycle, repeated until EoSL

Question to stakeholders:

- *Which of the three options do you consider more appropriate? Please provide arguments supporting your claim. Do you have any suggestions on how one or more options should be changed?*
- *Are you aware of any further standards that should be considered for the option 'application-specific standards' and that has to be added to Table 2? Please indicate also national standards that might be helpful for defining a consistent approach for determining service life*

The **reference flow** (RF) is the amount of product needed to fulfil the defined function and shall be measured in kg of battery per FU (i.e., per kWh of the total energy delivered by the battery over its service life). All quantitative input and output data collected by the manufacturer to quantify the carbon footprint shall be calculated in relation to this reference flow.

Correspondingly, the reference flow (RF) i.e., amount of battery mass required to provide one functional unit is calculated as the total mass of battery (mB) divided by the quantity of FU (QUa):

$$RF = m_B / QU_a$$

3.2 Functional unit for on-demand (OND) or backup industrial batteries

On-demand batteries (such as UPS systems) are only occasionally or even never cycled, being their principal function the ability to provide power for short times e.g., in case of power outage. On-demand batteries can provide their service either in mobile equipment (OND-MOB) or in stationary equipment (OND-STA).

The **functional unit** for the CFB of OND batteries is defined as:

"Provision of 1 kWmin of backup power capability over the whole service life. Provision of backup power refers to the ability to discharge at the given power without interruption at continuous discharge above minimum voltage at any time."

As for REP-batteries, different options are available for determining the service life.

Option 'manufacturer declares'

The **service life** shall be declared in terms of years until reaching the EoSL criterium and be provided in the CFB supporting study. it is defined as the Reference Service Life of the battery system for the application as described in the technical specification of the battery system as provided to the purchaser (i.e., the end-user). The corresponding use profile and reference service life specified for the battery or battery system within the specific application shall be provided in years together with the corresponding total energy and the reference test used in the CFB supporting study.

Option 'one standard for all'

The **service life** shall be declared in terms of years until reaching the EoSL criterium and be provided in the CFB supporting study. Unless specific standards are referenced in the EU's Official Journal for the purposes of Article 10 and Annex IV of the Batteries Regulation, the EoSL criteria shall be determined according to IEC 60896-21/22:2004 (details in Table 1).

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Option 'application-specific standards'

The **service life** shall be declared in terms of years until reaching the EoSL criterium and be provided in the CFB supporting study. Unless specific standards are referenced in the EU's Official Journal for the purposes of Article 10 and Annex IV of the Batteries Regulation, the EoSL criteria and test standards in Table 2 apply (Service = OND in Table). The EoSL criterium shall be indicated together with the declared service life. By default, the EoSL shall be set as the point when the battery has reached 80% of its declared rated energy capacity.

The provided **backup power capability** is determined by the rated power capability of the battery in kW (Pr) and its stored energy time (Tse) in minutes, available over the whole service life. Stored energy time (Tse) refers to the time (in minutes) the battery is able to provide power at Pr above the specified minimum voltage according to IEC 62040-3. It is equivalent to the rated energy capacity [kWh] / rated power [kW] * 60.

Correspondingly, the **quantity of functional unit** (QU_a); i.e., the amount of backup power capability (in kW·min) provided over service life by the battery can then be calculated by multiplying the service life in years (Ny) with the backup power capability (P_r * Tse):

$$QU_a = Ny * P_r * Tse$$

As for energy-providing batteries, the **reference flow** is measured in kg of battery per FU (i.e., per kW·min of backup power capability provided by the battery over its service life). All quantitative input and output data collected in the CFB study shall be calculated in relation to this reference flow.

The reference flow (RF) i.e., the amount of battery mass required in order to provide one functional unit is calculated as the total mass of battery (m_B) divided by the quantity of FU (QU_a):

$$RF = m_B / QU_a$$

Question to stakeholders:

- *Which of the options do you consider more appropriate? Please provide arguments supporting your claim. Do you have any suggestions on how one or more options should be changed?*
- *Are you aware of any further standards that should be considered for OND batteries?*

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4 System Boundaries and cut-off rules

The system boundaries define which parts of the product life-cycle and which associated life-cycle stages and processes belong to the analysed system (i.e., are required for carrying out its function as defined by the functional unit).

Batteries falling under the scope of this document are usually comprised of several battery cells grouped together forming a battery system. Such systems can have very diverse layouts and system configurations, given the wide range of applications they are designed for. Typically, several battery cells are grouped together to form battery modules, which are then mounted together in a specific structure to form a battery system such as a rack-based stationary battery or home storage system. Several of these systems in turn can be clustered and mounted into a common housing such as a container, forming a large stationary battery system. A schematic representation of the considered system components is provided in Figure 1. The definition of the system boundaries follows a general supply-chain logic, including all stages from raw material acquisition and pre-processing, production of the main product, product distribution, etc.

The following life-cycle stages shall be included in the system boundaries (Figure 2):

— 'Raw materials acquisition and pre-processing', described in section 4.2.

— 'Manufacturing' of the battery, described in section 4.3.

— 'Distribution', described in section 4.4.

— 'End-of-life', described in section 4.5.

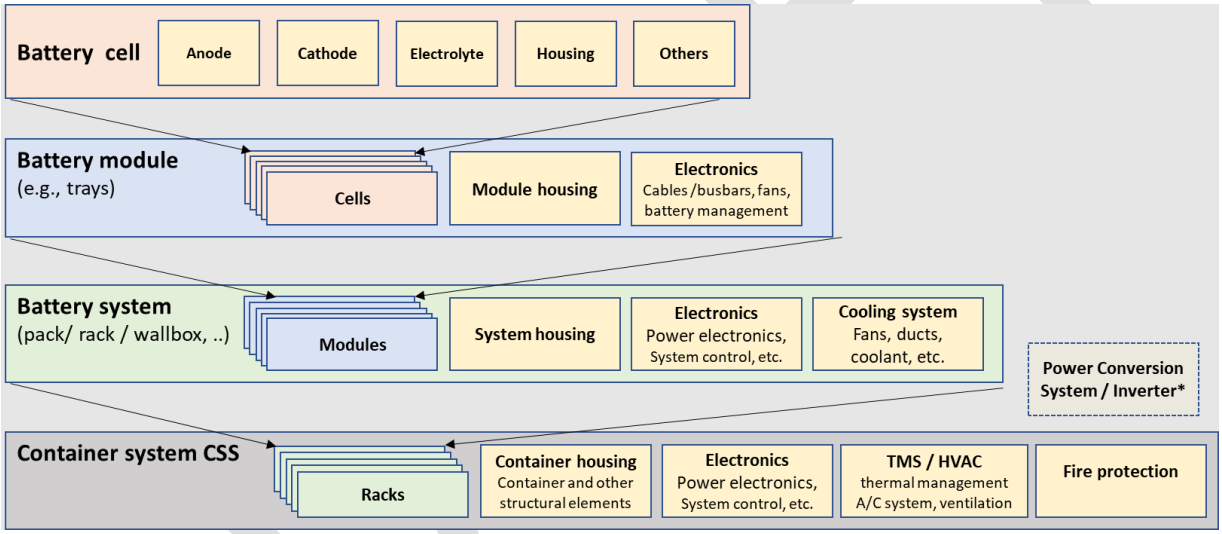


Figure 1. System components. * Power Conversion System / Inverter excluded from the considered system components. Based on the JRC report on CFB-EV.

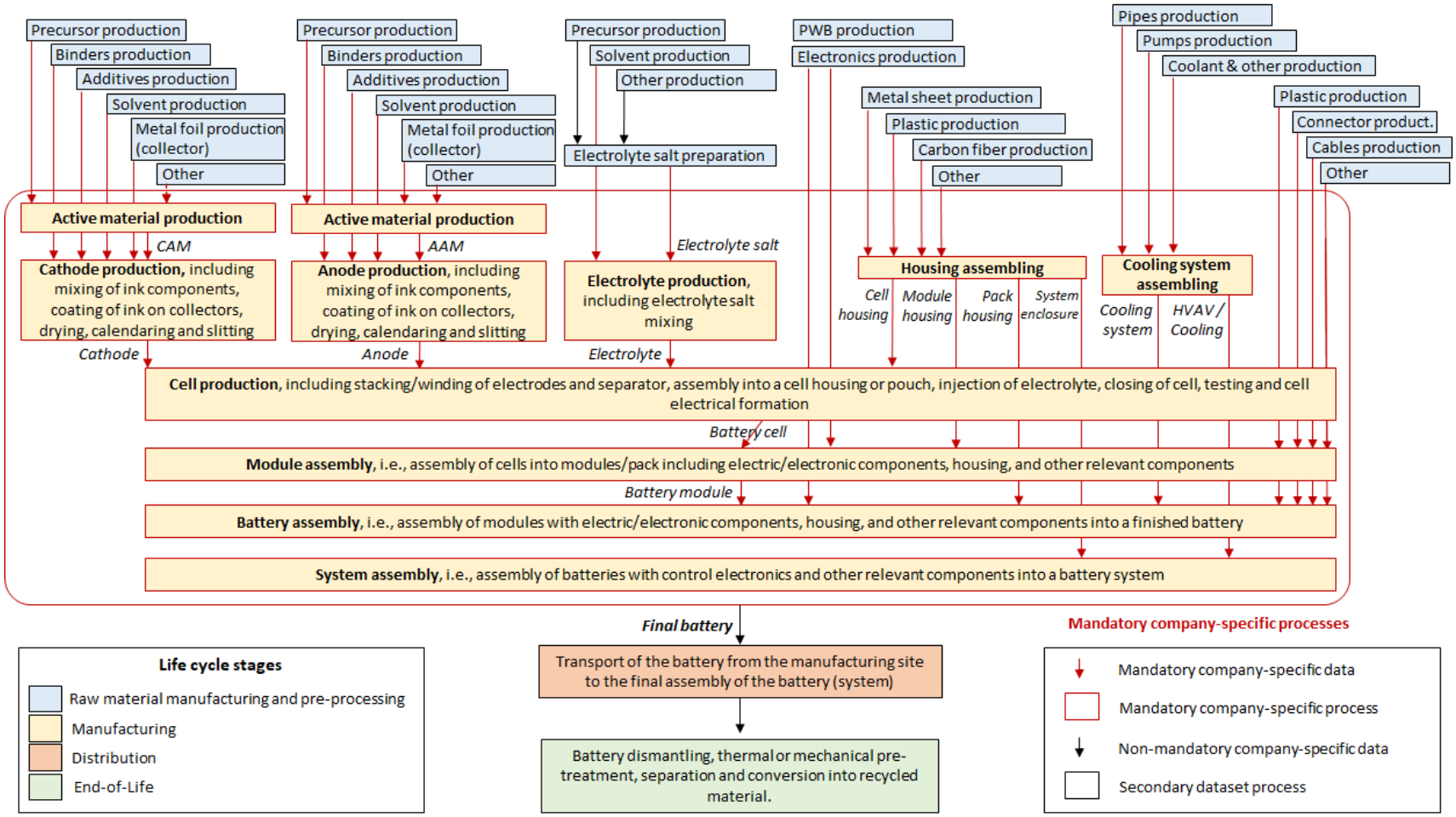


Figure 2. System boundaries of the carbon footprint of a generic industrial battery. Other types of batteries (e.g., anode-free batteries, cell-to-pack design) may have a different visualization of their system boundaries. Each square represents a process, while each arrow represents an activity data (e.g., kg of solvent, kg of additive). The different colours (blue, yellow, orange, and green) indicate to which life-cycle stage each process belongs, while red arrows and red borders indicates if a process/activity data shall be company-specific (section 6.1). Black arrows and borders indicate non-mandatory company-specific processes. PWB: Printed Wiring Board. CAM: cathode active material, AAM: anode active material

Question to stakeholders:

- Would you agree with the inverter being excluded from the system boundaries? Please comment both on OND and on REP batteries.
- What do you think about the inclusion (or exclusion) of the fire extinguishing system in the system boundaries if it is integrated (or not) inside the container?
- What about the charger to be considered as part of the battery energy storage system (e.g., for home storage battery)?

4.1 Cut-off rules

The following processes shall be excluded from the modelling:

— The use-phase according to Article 7 and Annex II of the Batteries Regulation.

— Manufacturing of equipment (capital goods).

— The installation of the finished battery in the final application environment.

— The power conversion system (PCS) to have a level basis for comparison, even if integral part of a battery system, being the PCS usually a separate component for the batteries in scope and being the actual output of a battery DC power.

— The production of packaging materials.

— Auxiliary inputs to the manufacturing plant that are not directly related to the battery production process (such as heating and lighting of associated office rooms, secondary services, sales processes, administrative and research departments, etc.)*.

— Storage operations during the 'Distribution' life-cycle stage.

— The impacts of the collection of the battery waste, the impacts of the pre-treatment of the waste batteries (e.g., on-site disassembly, sorting), the impacts of the dismantling of the battery and its component, and the transport during the 'End-of-life' life-cycle stage.

A general cut-off of 1% in mass may be applied to material inputs per system component as in Figure 1 (e.g., it is possible to neglect input and output flows of the housing that make up less than 1% to the total mass of the housing). If a cut-off is applied, the mass gap shall be closed on system component level by adding the missing mass to the input flow with the highest specific carbon footprint. The CFB-declarant shall report in the CFB supporting study if and where the cut-off of 1% in mass has been applied.

** NOTE: the auxiliary inputs to the 'Manufacturing' life-cycle stage were estimated to be negligible: assuming a consumption of a generic office as 293 kWh/m2 18, and assuming a production area of 267,000 m2 (compatible with a facility of 20 GWh/year of production capacity), the auxiliary energy consumption is estimated to be about 0.2% of the electrical energy required by the production line.*

4.2 Raw material acquisition and pre-processing

Identical to CFB-EV.

4.3 Manufacturing

The 'Manufacturing' life-cycle stage includes:

— Anode and cathode active material production.

— Anode and cathode production, including the mixing of ink components, coating of ink on collectors, drying, calendaring, and slitting.

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— Electrolyte production, including the electrolyte salt mixing.

— Cell production (i..e, the assembly of cell components into a battery cell), including stacking/winding of electrodes and separator, assembly into a cell housing or pouch, injection of electrolyte, injection of the electrolyte, closing of cell, testing and electrical formation.

— Module assembly, i.e., assembly of cells into modules/pack/rack including electric/electronic components, housing, and other relevant components.

— Battery assembly, i.e., assembly of modules with electric/electronic components, housing, and other relevant components into a finished battery

— Housing and cooling system assembly

— System assembly, i.e., transport of the batteries to the site of installation and assembly with the housing (e.g., container), power and control electronics, thermal management and fire suppression system and wiring.

— All transport operations of the final and intermediate products to the site where they are used.

More information on how to account for manufacturing scraps and recycled content in the 'Manufacturing' life-cycle stage is provided in section 7.3.

4.4 Distribution

The 'Distribution' life cycle stage considers the transport of the battery from the manufacturing site to the final installation of the battery within the application. For battery systems that are sold via retail channels (such as home storage systems), the following standard transport distances may be used, independently of the specific distribution channel (e.g., no distinction between distribution via retail stores or via delivery services):

From battery manufacturer to retail / client

For suppliers located within Europe (utilisation ratio 64%)

- 130 km by truck (28-32 t, EURO 5; 28-32 t, EURO 5), PEFCR specific utilisation ratio
- 240 km by train (average freight train)
- 270 km by ship (barge).

For all suppliers located outside Europe (utilisation ratio 64%)

- 1000 km by truck (>32 t, EURO 4), for the sum of distances from the harbour to customer outside and inside Europe. EF specific utilisation ratio; and
- 18000 km by ship (transoceanic container), 100% utilization rate

Question to stakeholders:

- Do you have any comment on how to model the 'Distribution' life cycle stage?

4.5 End-of-life

The 'End-of-life' life-cycle stage begins when the product in scope is disposed of or discarded by the user or the operator and ends when the product in scope is returned to nature as a waste product or enters another product's life-cycle (i.e., as a recycled input). On the other hand, waste generated during the 'Raw material acquisition and pre-processing' and 'Manufacturing' life-cycle stages shall be included in the life-cycle of the product and modelled at the life-cycle stage where it occurs. No refurbishing or second life shall be considered according to the Batteries Regulation i.e., a battery that reaches its end of life for the first time shall be considered to enter the End-of-Life stage.

The 'End-of-life' life-cycle stage includes:

— Battery dismantling

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- Thermal or mechanical pre-treatment (e.g., thermal deactivation, pyrolysis, shredding of the battery cell)
- Battery cell recycling (such as pyrometallurgical and hydrometallurgical treatment of the cells)
- Separation and conversion into recycled material
- Electronics (printed wiring board - PWB) recycling
- Energy recovery and disposal.

More information on how to model the 'End-of-life' life-cycle stage is provided in section 7.3

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5 Impact assessment

Identical to CFB-EV.

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6 Inventory data - Data collection requirements and quality requirements

A CFB model is composed of processes that combine elementary flows with the corresponding characterisation factors, and activity data with the corresponding life-cycle inventory or carbon footprint of the underlying process.

Both data (i.e., activity data and elementary flows) from processes and the underlying sub-processes can be company-specific or secondary (i.e., taken from databases). The CFB Rules distinguish between mandatory company-specific processes (section 6.1) and non-mandatory company-specific processes (section 6.2).

All data shall be provided as LCI and activity data (i.e., before and after allocation), both per kg of the main output (product) of the modelled process and per functional unit.

6.1 Mandatory company-specific processes

Identical to CFB-EV.

6.2 Non-mandatory company-specific processes

Non-mandatory processes are divided into most relevant and non-most relevant processes depending on their relevance on the overall CFB calculation.

The most relevant processes are the following:

- Production of cathode active material precursors: cobalt, nickel, iron, lithium, cadmium, lead, vanadium, rare earths (metallic or as salts).
- Production of anode active material precursors: lithium, titanium, nickel, lead (all both in metallic form or as salts), sodium metal, silicon, graphite, hard carbons,
- Production of electrolyte (salt) and precursors: $LiPF_6$, LiTFSI, lithium salts, nickel salts, solid electrolytes
- Production of copper, e.g., in the current collectors, busbars and cables.
- Production of aluminium, e.g., in the current collectors, busbars / cables and housing.
- Production of steel, e.g., in the housing.

NOTE: the most relevant processes include all that had been identified for EV batteries, plus several additional materials that have been deemed relevant for Industrial Batteries based on scientific literature and expert judgement. Examples of non-most relevant processes are the production of the solvent used in cathode and anode, of the carbon fibers used in the housing, of the PWB, the manufacturing of the plastic for the collector or housing, etc.

Question to stakeholders:

- Do you agree with the processes identified above as most-relevant for all types of battery chemistries in scope? Which processes would you add and/or remove?

6.2.1 Modelling requirements of the most relevant processes

Identical to CFB-EV.

6.2.2 Modelling requirements for the non- most relevant processes

Identical to CFB-EV.

6.3 CFB-compliant company-specific datasets

Identical to CFB-EV.

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6.4 Requirements for the collection of company-specific data

Identical to CFB-EV.

6.5 Data quality rating (DQR) of the CFB declaration and of CFB-compliant datasets

Identical to CFB-EV.

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7 Inventory data - Modelling requirements

7.1 Electricity modelling

Identical to CFB-EV

7.2 Allocation rules

Virtual housing not applicable. Otherwise identical to CFB-EV.

Question to stakeholders:

- Is there any specific process for industrial batteries that should be considered for the allocation rules?

7.3 Recycling content and end-of-life modelling

The recycled content and the waste generated during all the life-cycle stages shall be modelled with the use of the Circular Footprint Formula (CFF) and shall be reported at the life-cycle stage where the waste management occurs.

The CFF shall not be used for run-around scrap since this is not considered manufacturing waste according to the Article 3 of the Batteries Regulation. However, emissions and process inputs associated with their processing shall be accounted for in the corresponding life-cycle stage.

7.3.1 The CFF applied to the battery in scope

The below equations describe how to use the CFF to the battery in scope. In particular, it should be noted that:

— The CFF shall be applied per material, unless specified differently in the description of the single parameters.

— For clarity, the "material" part of the CFF has been divided into four terms, related with different processes along the production and end-of-life stage:

- Impacts of using primary and secondary materials in the battery production, i.e., when the recycled content is different than zero.
- Impacts of producing secondary materials from the dismantling: steel and aluminium from the housing and copper from the cables.
- Impacts of producing secondary materials from the PWB recycling after the battery dismantling: copper, gold, and palladium.
- Impacts of producing secondary materials from the battery cell recycling: copper, nickel sulphate, and cobalt sulphate in the default battery cell recycling of LIB; lead and antimony from the default battery cell recycling of lead-acid batteries.

— The default recycled content of the materials in the battery scope shall be zero unless evidence of the traceability throughout the supply-chain is provided in the CFB supporting study.

— The modelling requirements for the 'End-of-life' life cycle stage depend on the battery chemistry of the battery in scope (i.e., the battery for which the CFB is developed and declared). For all batteries, it is assumed that only a fraction of the batteries is properly collected and sent to (manual) dismantling (Figure 3). The outputs of the dismantling are the battery cell and remaining battery components that are sent to individual recycling processes (steel and aluminium from the housing; copper, PWB for recovering gold, silver, copper and palladium, cables for recovering copper). The remaining fraction of the batteries is assumed to be not properly collected and treated (named as "non-collected" batteries in Figure 3) and assumed to be roughly dismantled (with a process similar to that for properly collected batteries above, but without carrying out the recycling of the PWB), whereas battery cells are assumed to be landfilled. The sorted plastic is sent to energy recovery and the rest to landfill. The dismantled battery cell is then sent to the default battery cell recycling process, which varies according to the battery cell chemistry:

1. The default battery cell recycling process for lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid

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electrolytes operating at room temperature such as sealed pouch, round cell or prismatic cells shall be modelled according to Table 2 and Table 3, assuming a combination of a pyrometallurgical and hydrometallurgical treatment (that recovers copper, nickel sulphate, and cobalt sulphate).

2. The default battery cell recycling process for lead-acid batteries shall be modelled according to Table 5, using default EF datasets

3. For any other battery chemistry, no established battery cell recycling processes yet exist and therefore no battery cell recycling shall be modelled. The battery cell is assumed to be sent to landfilling.

— A company-specific collection rate may be used only for the share of batteries covered by an ownership business models where the property of the battery stays with the manufacturer and higher collection rates may be proved.

— A company-specific battery cell recycling process may be used only for the share of the batteries for which the CFB declarant is able to provide evidence that the same batteries will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the CFB is declared. The corresponding evidence shall be provided in the CFB supporting study.

— The impacts of collecting and dismantling the waste batteries shall not be included in the model (i.e., they are defined as cut-off in section 4.1). The collection rate is expressed with the parameters R_{coll} in the equation below, while the efficiency of the dismantling is assumed to be 100%.

— The impacts of the battery cell recycling shall not be allocated per material but shall be allocated to the full battery using an average A per battery cell.

— The impacts of recycling of the PWB shall not be allocated per material but shall be allocated to the full PWB using an average A per the PWB.

— The incineration with energy recovery only applies to the plastic waste from the dismantling of the properly collected batteries. The energy recovery from plastics in the battery cells is instead accounted for by the default battery cell recycling process.

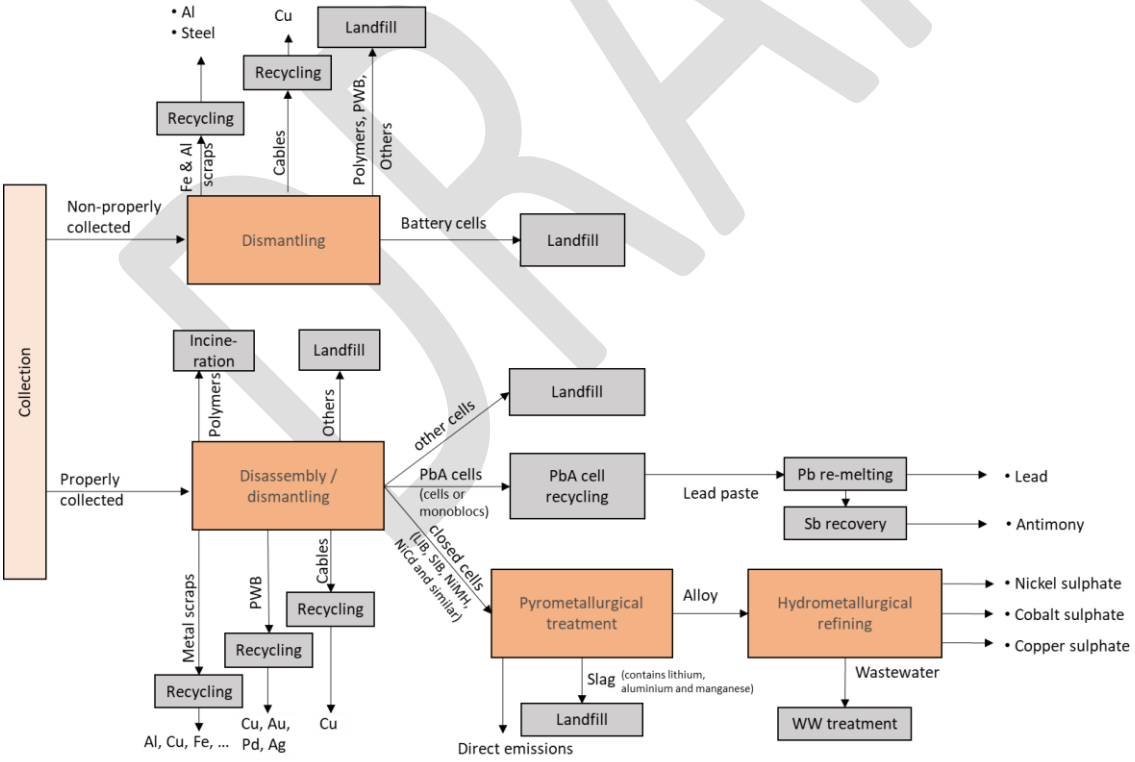


Figure 3. Schematic flowsheet of the default 'End-of-life' life-cycle stage of the battery in scope. PWB: printed wiring board

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Equations describing how to apply the CFF to the recycled content and to the 'End-of-life' life-cycle stage of the battery in scope are:

— (Material input): Impacts of using primary and secondary materials:

$$\sum_{Mat}[(1-R_{1_Mat)}xE_{v_Mat}+R_{1_Mat}x\left(A_{Mat}\ x\ E_{recycled_Mat}+(1-A_{Mat})\ x\ E_{V_Mat}\ x\ \frac{Q_{Sin_Mat}}{Q_{P_Mat}}\right)]$$

— (Dismantling): Impacts and credits of producing secondary materials from the dismantling (to calculate for steel and aluminium from the housing; copper from the cables):

  — Materials from the properly collected fraction of the battery

$$R_{coll}\ x\ \sum_{Mat}\left[(1-A_{Mat})\ x\ R_{rec,c_Mat}\ x\ \left(E_{recEoL_Mat}-E^{*}_{V_Mat}\ x\ \frac{Q_{Sout_Mat}}{Q_{P_Mat}}\right)\right]$$

  — Materials from the non-properly collected fraction of the battery

$$+(1-R_{coll})\ x\ \sum_{Mat}\left[(1-A_{Mat})\ x\ R_{rec,nc_Mat}\ x\ \left(E_{recEoL_Mat}-E^{*}_{V_Mat}\ x\ \frac{Q_{Sout_Mat}}{Q_{P_Mat}}\right)\right]$$

— (Electronics recycling) Impacts and credits of producing secondary materials from the PWB recycling after the battery dismantling (to calculate for copper, gold, silver, and palladium):

  — PWB from the properly collected fraction of the battery

$$R_{coll}\ x\ [(1-A_{PWB})\ x\ E_{recEoL_PWB}-\sum_{Mat}\left[(1-A_{Mat})\ x\ \left(R_{rec,c_Mat}x\ E^{*}_{V_Mat}\ x\ \frac{Q_{Sout_Mat}}{Q_{P_Mat}}\right)\right]$$

— (Cell recycling): Impacts and credits of producing secondary materials from the battery cell recycling (to be calculated for copper, nickel sulphate, and cobalt sulphate in the default 'End-of-life' life-cycle stage):

$$R_{coll}\ x\ \left((1-A_{Battery\ cell})\ x\ E_{recEoL_{Battery\ cell}}\right)$$

$$+R_{coll}\ x\ \sum_{Mat}\left((1-A_{Mat})\ x\ R_{rec,c_Mat}\ x\ \left(E_{recEoL_Mat}-E^{*}_{V_Mat}\ x\ \frac{Q_{Sout_Mat}}{Q_{P_Mat}}\right)\right)$$

— (Energy recovery): Impacts due to energy recovery of the dismantled plastic:

$$R_{coll}\ x\ \sum_{Mat}[(1-B)\ x\ R_{3,c_Mat}\ x\ (E_{ER_{Mat}})]$$

— (Disposal): Impacts of disposal:

  — PWB, polymers, battery cell, and other materials not-being recycled, deriving from the dismantling of the non-properly collected battery waste

$$(1-R_{coll})\ x\ \sum_{Mat}(1-R_{rec,nc_Mat})\ x\ E_{D_{Mat}})$$

  — Other materials not-being recycled, deriving from the dismantling of the properly collected battery waste

$$R_{coll}\ x\ \sum_{Mat}(1-R_{rec,c_Mat}-\ R_{3,c_{Mat}})\ x\ E_{D_{Mat}})$$

All the above equations shall be summed to obtain the final result.

The parameters of the CFF to be applied to the battery recycled content and to the battery 'End-of-life' life-cycle stage[3] are:

— A_{Mat}: Material-specific allocation factor of burdens and credits between two life-cycles (i.e. the one supplying and the one using recycled material) aiming to reflect market realities. Values of parameter "A_{Mat}" shall be selected according to Table 6.

[3] If the CFF is to be applied to model the waste treatment in other life-cycle stages, the CFF and the CFF parameters as described in the section 4.4.8 of the general EF method apply.

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- $A_{Battery\ cell}$: Battery cell-specific allocation factor. It shall be equal to 0.2.
- A_{PWB}: PWB-specific allocation factor. It shall be equal to 0.2.
- B: allocation factor of energy recovery processes. It shall be equal to 0.
- R_{1_Mat}: Material-specific recycled content, i.e., the proportion of material in the input to the production that has been recycled from a previous system. R_1 is by default equal to 0 for all the materials, unless a different value may be duly proved. When using a company-specific R_1 value other than 0, traceability throughout the supply-chain is necessary, and evidence shall be provided in the CFB supporting study for verification. The documentation required to demonstrate the minimum recycled content according to the Article 8 of the Battery Regulation are considered evidence also for an R1 different than 0. Material-specific values based on supply market statistics shall not be accepted as a proxy for R_1.
- R_{coll}: Battery collection rate (i.e. share of the end-of-life batteries that are properly collected to be dismantled and recycled). The default R_{coll} is 0.7 for all batteries. Different company-specific values may only be used for the share of batteries covered by an ownership business models where the property of the battery stays with the manufacturer and higher collection rates may be proved. Evidence of such ownership models shall be provided in the CFB supporting study.

NOTE: The collection rate represents the percentage of industrial batteries that are not treated in a battery cell recycling process.

- R_{rec,c_Mat} and R_{rec,nc_Mat}: Material-specific recycling yield, i.e., the proportion of the material in the product that will be recycled in a subsequent system taking into consideration only the recycling yield. It does not include collection rate and dismantling efficiency. Two different values are used, R_{rec,c_Mat} for the properly collected fraction of batteries and $R_{rec,nc,Mat}$ for the non-properly collected fraction. R_{rec,c_Mat} and R_{rec,nc_Mat} values of the default recycling process are provided in Table 6 for all lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature such as sealed pouch, round cell or prismatic cells, and for all lead-acid batteries. For all other battery chemistries, the default R_{rec,c_Mat} for the cell recycling process is zero. Company-specific R_{rec,c_Mat} values may be used only for the share of the batteries for which the CFB declarant is able to provide evidence that the same batteries will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the CFB is declared. In case of company-specific R_{rec,c_Mat} values, they shall be calculated as the ratio between the mass of the secondary material produced and the mass of the material entering the recycling process. The corresponding evidence (such as contractual instruments and the number of batteries being recycled under them) shall be provided in the CFB supporting study. Even in the case of company-specific R_{rec,c_Mat}, these may be applied only for the amount of batteries properly collected. In the case of the material recycled from the PWB recycling (copper, gold, silver, and palladium), R_{rec,c_Mat} refers to the kg of material per kg of PWB in input and only the default values provided in Table 6 shall be used.
- R_{3,c_Mat}: the proportion of the material that is used for energy recovery at the end-of-life of the properly collected waste batteries and materials. In the default 'End-of-life' life-cycle stage, only the dismantled polymers from the dismantling goes to energy recovery (i.e., R_{3,c_Mat} = 100% for polymers). For non-properly collected waste batteries, no energy recovery is assumed, so the proportion of the material in the product that is used for energy recovery at the end-of-life is equal to 0 by default. For other materials, $R_{3,c}$ shall be 0 unless proven in the carbon footprint supporting study._
- E_{V_Mat}: specific emissions and resources consumed arising from the acquisition and pre-processing of primary material. E_V may be either a most relevant process or a non-most relevant process (section 6.2).
- $E_{recycled_Mat}$: specific emissions and resources consumed arising from the recycling process of the recycled material. Collection, sorting, and transportation of the waste used to produce the secondary material shall be excluded. $E_{recycled_Mat}$ can be either a most relevant process or a non-most relevant process (section 6.2). The allocation of a multi-material process to the single material shall follow the hierarchy for dealing with multifunctionality described in section 7.2. It shall include the management of the waste from the recycling process and the wastewater treatment (when applicable).
- $E_{recEoL_Battery\ cell}$: specific emissions and resources consumed arising from the battery cell recycling. $E_{recEoL_Battery\ cell}$ is accounted for on battery level and not per material. The default battery cell recycling process is assumed to take place in Europe. Different company-specific recycling processes may be used only with

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the conditions described for the R_{rec,c_Mat}. In this case, the recyclers may: i) provide all the data, ii) provide a CFB-compliant company-specific dataset to the CFB declarant. Three different default $E_{recEoL_Battery\ cell}$ are identified:

- For lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature such as sealed pouch, round cell or prismatic cells, the default battery cell recycling process as provided in Table 2 and Table 3 shall be used.
- For lead-acid battery cells, the battery cell recycling shall be modelled using the default recycling process as provided in Table 5.
- For all other battery chemistries, e.g., sodium high temperature batteries, the battery cell is assumed to be sent to landfill.

— E_{recEoL_Mat}: specific emissions and resources consumed arising from any additional recycling process that is needed to produce secondary material. For all the outputs of the dismantling, the default recycling processes are assumed to take place in Europe both for the properly collected waste batteries and the non-properly collected waste batteries (e.g., sorting and re-melting of aluminium waste scraps of the casing into secondary aluminium). For all the outputs of the battery cell recycling process, E_{recEoL_Mat} includes all the additional recycling processes not included in the battery cell recycling. It is equal to zero in the default battery cell recycling process since the obtained products do not require further post-processing.

NOTE: $E_{recEoL_Battery\ cell}$ includes all process inputs and direct emissions of the battery cell recycling process. E_{recEoL_Mat} are the specific burdens (carbon footprint) associated with the re-processing of the metal fractions separated during recycling, such as sorting and re-melting of aluminium scrap into secondary aluminium. E_{recEoL_Mat} is zero for all materials that leave the battery cell recycling process as final product i.e., directly replacing the equivalent primary product such as cobalt sulphate or nickel sulphate as in the default EoL life cycle stage. $E_{recEoL_{Mat}}$ is not zero for any output of the battery cell recycling process that needs additional processes before being sold as secondary material. (e.g., sorting, cleaning and remelting to metal ingots).

— E_{recEoL_PWB}: specific emissions and resources consumed arising from the PWB recycling following the dismantling of the properly collected batteries. E_{recEoL_PWB} shall always be modelled as a non-most relevant process (section 6.2.2).

— $E^*_{v_Mat}$: specific emissions and resources consumed arising from the acquisition and pre-processing of primary material assumed to be substituted by recyclable materials. Ev* shall be equal to Ev.

— Q_{sin}: quality of the ingoing secondary material, i.e., the quality of the recycled material at the point of substitution.

— Q_{Sout}: quality of the outgoing secondary material, i.e., the quality of the recyclable material at the point of substitution.

— Q_p: quality of the primary material.

— Q_{sin} / Q_p: ratio between the quality of the ingoing secondary material and the quality of the primary material. Default values are provided in Table 6.

— Q_{Sout} / Q_p: ratio between the quality of the outgoing secondary material and the quality of the primary material. Default values are provided in Table 6. Higher values are allowed only if a company-specific battery recycling process is modelled and if justified in the CFB supporting study, providing e.g., the technical specifications of the materials demonstrating that is battery-grade.

— E_{ER}: specific emissions and resources consumed arising from the energy recovery of the polymers from the manual dismantling (for both the properly collected and the non-properly collected waste batteries). E_{ER} shall always be modelled as a non-most relevant process (section 6.2.2). If the secondary datasets for energy recovery do not include the credits from the production of energy, the CFB declarant shall model the credits using the lower heating value of the polymers.

– E_D: specific emissions and resources consumed arising from the disposal of waste material at the analysed product's end-of-life, without energy recovery.

For PWB, the CFB declarant shall check if the datasets for PWB recycling already include the credits from the production of secondary material to avoid double accounting. If the credit for recovered materials is already contained in the LCI data or the LCIA result, the credits for specific materials shall be set to zero. For

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$E_{recycled_Mat}$, the CFB declarant shall check that the datasets did not already allocated the impacts with the factor A to avoid double allocation of the impacts.

To apply the CFF correctly along the battery supply-chain, the company-specific R1 of e.g., any raw material, precursors, active material, shall be handed over to the next process step / manufacturer even in case of CFB-compliant company-specific datasets.

Table 3. Default cell recycling model for recycling of LIB, NiMH, SIB and other battery cells of similar layout, inventory data per kg of battery cell, pyrometallurgical treatment. Equal to the CFB-EV.

Component	Unit	Default value	Dataset	Specification
Input				
EoL battery cells	kg	1.00	From disassembly	
Electricity	kWh	1.00	Residual electricity mix	EU residual grid mix
Process heat	MJ	2.288	Thermal energy from natural gas	
Process heat	MJ	0.237	Thermal energy from diesel fuel	
Limestone washed	kg	0.136	Lime production, technology mix	
Silica sand	kg	0.119	Silica sand, production mix	
Quicklime (CAO)	kg	0.085	Quicklime production, technology mix	
Carbon black	kg	0.001	Carbon black production, technology mix	
Outputs				
Metal alloy	kg	0.209	To hydrometallurgical treatment	May be adjusted according to specific battery cell composition (mass of cells fed to treatment - mass of alloy - mass of oxidised components accounted for in the direct CO2 emissions)
Slag	kg	0.712	Landfill of inert slag	Fixed value per kg of cell (shall not be adjusted acc. to specific cell composition)
Emissions	kg CO2eq	1.194	Direct process emissions (all greenhouse gases)	Shall be adjusted according to specific battery cell composition (estimated based on stoichiometric calculation, assuming complete oxidation of all organic or carbonaceous compounds during pyrometallurgy)

Source: JRC analysis.

Table 4. Default recycling model for recycling of LIB, NiMH, SIB and other battery cells of similar layout, inventory data per kg of battery cell, hydrometallurgical treatment.

Component	Unit	Default value	Dataset	Specification
Input				
Metal alloy	kg	0.209	--	Fixed value per kg of cell
Electricity	kWh	2.59	Residual electricity mix	EU residual grid mix
Process heat	MJ	4,78	Thermal energy from natural gas	

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Hydrochloric acid (100%)	kg	0.011	Hydrochloric acid production, technology mix	
Hydrogen peroxide (100%)	kg	0.274	Hydrogen peroxide production, technology mix	
Soda (sodium carbonate)	kg	0.017	Soda production, technology mix	
Sodium hydroxide (100%; caustic soda)	kg	1,395	Sodium hydroxide production, technology mix	
Sulphuric acid aq. (96%)	kg	0.821	Sulphuric acid production, technology mix	
Water (tap water)	kg	10,41		
Outputs				
Recovered metals	kg		e.g., Aluminium, Copper, steel, stainless steel, ..	Credit for each specific metal according to its content in the battery cell and parameters Rcoll, A, Rrec and 'Qsout/Qp' of the CFF
Recovered metal salts	kg		e.g., Cobalt sulphate, Nickel sulphate ..	Credit for each specific metal salt according to its stoichiometric content in the battery cell and parameters Rcoll, A, Rrec, and 'Qsout/Qp' of the CFF
Wastewater	kg	8,64	Wastewater treatment, average	

Source: JRC analysis.

Table 5. Default recycling model for recycling of PbA battery cells or monoblocs, inventory data per kg of battery cell. The table is deliberately left empty due to insufficient availability of default data.

Component	**Unit**	**Default value**	**Dataset**	**Specification**
EoL PbA battery	**kg**	**1**	**--**	
Electricity	kWh		Residual electricity mix	EU residual grid mix
Process heat	MJ		Thermal energy from natural gas	
Outputs				
Recovered metals	kg		e.g., lead, antimony.	Credit for each specific metal according to its content in the battery cell and parameters Rcoll, A, Rrec and 'Qsout/Qp' of the CFF
Wastewater	kg	8,64	Wastewater treatment, average	

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Table 6. Default values for the parameters to be used in the default 'end-of-life' life cycle stage. [a] Company-specific values may be used according to section 7.3.1. [b]These values are to be applied per kg of PWB, since they indicate the mass of each material recovered per kg of recycled PWB. They shall match the corresponding recovered material flows provided in the used dataset. If credits are already accounted for within the dataset and no material output is given, the corresponding Rrec,c values zero. [c] Additional materials may be recycled if a company-specific battery cell recycling process is modelled. [d] Default values for for lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature such as sealed pouch, round cell or prismatic cells, and for lead-acid battery cells. [e] the recycling can only be claimed for lead and nickel not in the battery cell. For all other cell chemistries, the values for R_{rec,c_Mat} are zero.

	Material input			From dismantling / disassembly		From cell recycling [a, d]	
	A_{Mat}	R_{1_Mat}	Q_{sin}/Q_p	R_{rec,c_Mat}	$Q_{Sout,c}/Q_p$	R_{rec,c_Mat}	$Q_{Sout,c}/Q_p$
Metals and metal salts							
Aluminium	0.2	0 [a]	1	0.9	1	0	1
Antimony	0.2	0	1	--	--	0.9	1
Copper	0.2	0 [a]	1	0.9	1	0.9	1
Cobalt	0.2	0 [a]	1	--	--	0.9 [a]	0.8 [a]
Iron / Steel	0.2	0 [a]	1	0.9	1	0	1
Lead	0.2	0 [a]	1	0.95[e]	1	0.95	1
Lithium	0.2	0 [a]	1	--	--	0 [a]	0.8 [a]
Manganese	0.2	0 [a]	1	--	--	0 [a]	0.8 [a]
Nickel	0.2	0 [a]	1	0.9[e]	0.8	0.9 [a]	0.8 [a]
Tin	0.2	0	1			0.9	1
Other metals and metal salts	0.2	0 [a]	1	0	1	0 [a]	0.8 [a]
Electronics / PWB							
PWB/electr.: Copper	0.2	0 [a]	1	0.11[b]	1	--	--
PWB/electr.: Gold	0.2	0 [a]	1	1.40E-5[b]	1	--	--
PWB/electr.: Silver	0.2	0 [a]	1	9.77E-4[b]	1	--	--
PWB/electr.: Palladium	0.2	0 [a]	1	9.31E-8[b]	1	--	--
Other materials							
Polymers	0.5	0 [a]	1	0	0.8	0	0.8
Other materials [c]	0.5	0 [a]	1	0	--	0 [c]	0.8 [c]
(AAM) Graphite / hard carbon (cell)	0.2	0 [a]	1	--	--	0 [a]	0.8 [a]

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Source: JRC analysis.

Question to stakeholders:

- *Would you agree with the modelling approach and the proposed default values? If not, please suggest alternatives together with the corresponding data or evidence.*
- *Should we have different collection rates depending on the type of industrial battery?*
- *Would you agree with the modelling approach suggested for PbA Batteries and for 'other' batteries? If not, please provide an alternative proposal, e.g., the default battery cell recycling process to be used, including the corresponding evidences.*
- *Do you agree on the default CFF parameters as defined in the Tables above?*
- *Would you be able to provide data for PbA recycling process (no reliable literature source available))? And for other batteries that are not covered by Tables 3-5?*

7.3.2 The CFF applied to manufacturing waste

Identical to CFB-EV.

7.4 Transport

Identical to CFB-EV.

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8 Verification

8.1 Application for verification

Identical to EV batteries

8.2 Technical documentation

8.2.1 Public version of the CFB supporting study

The public version of the CFB supporting study shall at least contain:

- An unambiguous description of the product for which the CFB declaration is valid, including its origin and a unique identifier.
- Reference year (for which the CFB declaration was elaborated).
- The total CFB declared per FU (kg CO_2 eq/kWh) with a resolution of 0.001 kg CO_2 eq.
- The DQR score and the values of the single DQR criteria calculated as in section 6.5.
- The contribution of each life-cycle stage ('Raw materials and pre-processing', 'Manufacturing', 'Distribution', and 'End-of-life') to the total CFB of the battery per FU (in kg CO2eq/kWh).
- The CFB per kWh of rated energy capacity (kg CO_2eq/kWh).
- The rated energy capacity of the battery in kWh.
- The service life used for the calculation of the total energy provided over lifetime, expressed in number of full cycle equivalents and the cycle life test used for determining the service life (for energy-providing / repetitive energy supply (REP) batteries)
- The service life in years and the backup power capability in kWmin acc. to Section 3.2 over lifetime, and the reference test used (for backup / on-demand (OND) batteries). The end-of-life SOCE or remaining useable energy capacity of the service life test.
- The total energy (total energy provided by the battery over its service life as used for the CFB calculation) in kWh.
- The average discharge depth during each cycle of the cycle life test and the average amount of delivered energy over each cycle.

8.3 Verification and validation techniques

Identical to EV batteries

8.4 Data confidentiality

Identical to EV batteries

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9 Conclusions

The present report proposes a series of (draft) rules for determining the carbon footprint of industrial batteries with a capacity above 2 kWh, excluding those with exclusively external storage, in line with the requirements of the EU Batteries Regulation. It is based on the CFB Rules for EV batteries as published by the JRC in June 2023, and follows a delta approach, with only those Sections developed in detail where relevant discrepancies are given. All remaining Sections are assumed to remain identical to those of the CFB Rules for EV (as previously developed by JRC in June 2023)

This document is intended to provide the basis for exchange and stakeholder feedback towards a final report on the CFB Rules for Industrial Batteries (foreseen for Q2/2024) and for the corresponding Delegated Act as under the Article 7 of the Batteries Regulation.

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List of abbreviations

AC	Alternating current
ASIB	Aqueous Sodium-Ion Battery
BMS	Battery Management System
CFB	Carbon Footprint of Batteries
CFF	Circular Footprint Formula
CSS	Container Storage System
DC	Direct Current
EF	Environmental Footprint
E/P	Energy-to-Power ratio i.e., the ratio between energy and power rating (kWh/kW)
EMS	Energy Management System
EoL	End Of Life
EV	Electric Vehicle
FCE	Full Cycle Equivalents
FU	Functional Unit
GWP	Global Warming Potential
LCO	Lithium Cobalt Oxide (cathode material)
LFP	Lithium Iron Phosphate (cathode material)
LIB	Lithium-Ion Battery
LMO	Lithium Cobalt Oxide (cathode material)
LTO	Lithium Titanate Oxide (anode material)
Na-S	Sodium-Sulfur (battery)
Na-NiCl	Sodium-Nickel Chloride (battery), also known as ZEBRA battery
NCA	Lithium Nickel Cobalt Aluminium Oxide (cathode material)
NiCd	Nickel-Cadmium (battery)
NiMH	Nickel Metal Hydride (battery)
NiZn	Nickel-Zinc (battery)
NMC	Lithium Nickel Manganese Cobalt Oxide (cathode material)
PbA	Lead Acid (battery)
PCS	Power Conversion System
PEF	Product Environmental Footprint
PEFCR	Product Environmental Footprint Category Rules
PR	Primary Regulation
PV	Photovoltaics
PWB	Printed Wiring Board
RF	Reference Flow
SBS	Stationary Battery System
SIB	Sodium-Ion Battery
TMS	Thermal Management System

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UPS	Uninterruptible Power Supply

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List of definitions

'Accreditation' means accreditation as defined in Article 2(10) of Regulation (EC) No 765/2008 (Article 2 of the Battery Regulation Proposal).

'Active Material' refers to the materials within the battery material which react chemically to produce electric energy when the battery cell discharges or to store electric energy when the battery is being charged. They are the main constituents of the anode and cathode, respectively. For lithium-ion batteries, anode active materials are typically graphites, sometimes doped with silicon, or spinels such as lithium titanate. Cathode active materials (CAM) are typically layered oxides such as NMC or NCA, but also polyanionic materials such as LFP

'Active Material Precursor' are materials required for the synthesis of active materials, such as lithium salts (carbonates or hydroxides) and cobalt, nickel and manganese salts (most commonly sulphates) for the synthesis of NMC active materials

'Activity data' means the information associated with processes while modelling Life-cycle Inventories (LCI). The aggregated LCI results of the process chains that represent the activities of a process are each multiplied by the corresponding activity data and then combined to derive the carbon footprint associated with that process (Annex II of the Battery Regulation Proposal).

'Aggregated dataset' means life-cycle inventory (LCI) of multiple unit processes (e.g. material or energy production) or life-cycle stages (cradle-to-gate), but for which the inputs and outputs are provided only at the aggregated level. NOTE: Aggregated datasets are also called "LCI results", "cumulative inventory" or "system processes" datasets.

'Battery' means any device delivering electrical energy generated by direct conversion of chemical energy, having internal or external storage, and consisting of one or more non-rechargeable or rechargeable battery cells, modules or of packs of them, including a battery that has been subject to preparing for re-use, preparing for repurpose or repurposing, or remanufacturing (Article 2 of the Battery Regulation Proposal).

'Battery cell' means the basic functional unit in a battery constituted by electrodes, electrolyte, container, terminals and, if applicable, separators, and containing the active materials the reaction of which generates electrical energy (Article 2 of the Battery Regulation Proposal).

'Battery management system' (BMS) means an electronic device that controls or manages the electric and thermal functions of the battery in order to ensure the battery's safety, performance and service life, that manages and stores the data on the parameters for determining the state of health and expected lifetime of batteries laid down in Annex VII and that communicates with the vehicle, light mean of transport or appliance in which the battery is incorporated, or with a public or private charging infrastructure (Article 2 of the Battery Regulation Proposal).

'Battery manufacturing waste' means the materials or objects rejected during the battery manufacturing process, which cannot be re-used as an integral part in the same process and need to be recycled (Article 2 of the Battery Regulation Proposal).

'Battery module' means a set of battery cells that are connected together or encapsulated within an outer housing to protect the cells against external impact, and which is meant to be used either stand-alone or in combination with other modules (Article 2 of the Battery Regulation Proposal).

'Battery pack' means any set of battery cells or modules that are connected together or encapsulated within an outer housing, so as to form a complete unit that the end-user is not intended to split up or open (Article 2 of the Battery Regulation Proposal).

'Battery system' refers to batteries of any configuration subject to a mandatory CFB declaration under the Article 7 of the Battery Regulation Proposal. In this report, the term 'battery' is used as equivalent.

'Battery due diligence' means the obligations of the economic operator, in relation to its management system, risk management, third party verifications and surveillance by notified bodies and disclosure of information with a view to identifying, preventing and addressing actual and potential social and environmental risks linked to the sourcing, processing and trading of the raw materials and secondary raw materials required for battery manufacturing including suppliers in the chain and their subsidiaries or subcontractors that perform such activities (Article 2 of the Battery Regulation Proposal).

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'Battery model' means a version of a battery of which all units share the same technical characteristics relevant for sustainability and safety requirements and labelling, marking and information requirements pursuant to this Regulation and the same model identifier (Article 2 of the Battery Regulation Proposal).

'Bill of materials' means list of the raw materials, sub-assemblies, intermediate assemblies, sub-components, parts and the quantities of each needed to manufacture the product in scope of the study (Battery Regulation Proposal - Annex II).

'Carbon footprint' means the sum of greenhouse gas (GHG) emissions and GHG removals in a product system, expressed as carbon dioxide (CO2) equivalents and based on a Product Environmental Footprint (PEF) study using the single impact category of climate change (Article 2 of the Battery Regulation Proposal).

'CFB declarant' means the legal subject/entity that has the obligation to declare the CFB according to the Battery Regulation Proposal.

'CFB-compliant company-specific dataset' means a company-specific dataset that respects the requirements defined in section 6.3. A CFB-compliant company-specific dataset can be either a LCI dataset or a LCIA dataset

'Characterisation factor' means a factor derived from a characterisation model which is applied to convert an assigned life-cycle inventory result to the common unit of a LCIA impact category.

'Circular Footprint Formula' describes how burdens and benefits from disposal and recovery of the product or service assessed as well as use of secondary materials (i.e., recycled content) into that product or service are allocated to the system under study.

'Climate change' is the LCIA impact category considering all inputs and outputs that result in greenhouse gas (GHG) emissions. The consequences include increased average global temperatures and sudden regional climatic changes.

'Company-specific data' refers to directly measured or collected data from one or multiple facilities (site-specific data) that are representative for the activities of the company. It includes company-specific activity data and elementary flows. It is synonymous to 'primary data' (Annex II of the Battery Regulation Proposal) or 'supply-chain specific data' or 'manufacturer-specific' data.

'Company-specific dataset' means a dataset (disaggregated or aggregated) compiled with company-specific data. The actvity data and direct elementary exchanges of company-specific data sets shall refer to company-specific information, while the underlyin sub-processes may be modelled with company-specific or secondry datasets.

'Conformity assessment' means the process demonstrating whether the sustainability, safety, labelling, information or due diligence requirements of this Regulation have been fulfilled (Article 2 of the Battery Regulation Proposal).

'Conformity assessment body' means a body that performs conformity assessment activities including calibration, testing, certification and inspection (Article 2 of the Battery Regulation Proposal).

'Data' includes activity data and elementary flows.

'Data collection period' means the date(s) or time period(s) when the data was collected. Note that this does NOT refer to e.g., the publication dates of papers or books from which the data may stem, but to the original data collection period.

'Distributor' means any natural or legal person in the supply-chain, other than the manufacturer or the importer, who makes a battery available on the market (Article 2 of the Battery Regulation Proposal).

'Economic operator' means the manufacturer, the authorised representative, the importer, the distributor or the fulfilment service provider or any other natural or legal person who is subject to obligations in relation to manufacturing batteries, preparing batteries for reuse, preparing batteries for repurpose, repurposing, or remanufacturing, of batteries, making them available or placing them on the market, including on-line placing on the market, or putting them into service in accordance with this Regulation (Article 2 of the Battery Regulation Proposal).

'EF-compliant dataset' means dataset developed in compliance with the EF requirements in terms of modelling and methodological compliance (in agreement with the "Commission Recommendation (EU)

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2021/2279"); meta data compliance (in agreement with Fazio et al. 2020, "Guide on EF compliant datasets"); nomenclature, and characterization factors in agreement with the "EF 3.1 Reference Package"). The DQR of each single data quality indicator shall be lower or equal than 3.

'Elementary flows' or 'direct elementary flows' include 'material or energy entering the system being studied that has been drawn from the environment without previous human transformation, or material or energy leaving the system being studied that is released into the environment without subsequent human transformation'. Elementary flows include, for example, resources taken from nature or emissions into air, water, soil that are directly linked to the characterisation factors of the EF impact categories.

'Electric vehicle battery' or 'EV battery' means any battery specifically designed to provide electric power for the traction to hybrid or electric vehicles of L category as provided for in Regulation (EU) No 168/2013, and with a weight above 25 kg, or designed to provide electric power for the traction to hybrid or electric vehicles of M, N or O categories as provided for in Regulation (EU) 2018/858 (Article 2 of the Battery Regulation Proposal).

'Functional unit' means the qualitative and quantitative aspects of the function(s) and/or service(s) provided by the product being evaluated (Article 2 of the Battery Regulation Proposal).

'Harmonised standard' means a standard as defined in Article 2(1)(c) of Regulation (EU) No 1025/2012 (Article 2 of the Battery Regulation Proposal).

'ILCD entry-level compliant dataset (ILCD-EL)' means a dataset developed in agreement with the ISO 14040 and 14044 in terms of methodological compliance; in agreement with "International Reference Life-cycle Data System (ILCD) Data Network - Compliance rules and entry-level requirements" in terms of meta data compliance; in agreement with the "EF 3.1 Reference Package" for nomenclature and characterization factors compliance.

'Industrial battery' means any battery: - designed specifically for industrial uses, or – intended for industrial uses after being subject to preparing for repurpose or repurposing, or – any other battery with a weight above 5 kg that is not a LMT battery, an electric vehicle battery or a SLI battery (Article 2 of the Battery Regulation Proposal).

'Importer' means any natural or legal person established within the Union who places a battery on the market from a third country (Article 2 of the Battery Regulation Proposal).

'Independent operator' means a natural or legal person who is independent from the manufacturer and the producer and is directly or indirectly involved in the repair, maintenance or repurposing of batteries, and include waste management operators, repairers, manufacturers or distributors of repair equipment, tools or spare parts, as well as publishers of technical information, operators offering inspection and testing services, operators offering training for installers, manufacturers and repairers of equipment for alternative-fuel vehicles (Article 2 of the Battery Regulation Proposal).

'Intermediate product' means the output form of a unit process that in turn is input to other unit processes which require further transformation within the system. An intermediate product is a product that requires further processing before it is saleable to the final consumer

'Life-cycle' means the consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal (ISO 14040:2006 or equivalent) (Article 2 of the Battery Regulation Proposal).

'Life-cycle assessment (LCA)' means the compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life-cycle.

'Life-cycle inventory (LCI)' means the combined set of exchanges of elementary, waste and product flows in a LCI dataset.

'Life-cycle inventory (LCI) dataset' means a document or file with life-cycle information of a specified product or other reference (e.g., site, process), covering descriptive metadata and quantitative life-cycle inventory. A LCI dataset could be a unit process dataset, partially disaggregated or an aggregated dataset.

'Life-cycle inventory (LCI) results' means the quantitative assessment of the inputs and outputs associated with the life-cycle of a product, process, or system. LCI results provide detailed information about the environmental exchanges and resource consumption throughout the entire life-cycle, including the

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extraction of raw materials, manufacturing, transportation, use, and disposal. By following the supplies of intermediate inputs of each dataset and cumulating the environmental exchanges of these interlinked datasets.

'Life-cycle impact assessment (LCIA)' is a phase of the life-cycle assessment that aims to understand and evaluate the magnitude and significance of the potential environmental impacts for a system throughout the life-cycle. The LCIA methods provide impact characterisation factors for elementary flows to aggregate the impact, to obtain a limited number of midpoint and/or damage indicators.

'Life-cycle impact assessment (LCIA) dataset' means a document or file with life-cycle information of a specified product or other reference (e.g., site, process), covering descriptive metadata and quantitative life-cycle impact assessment results.

'Life-cycle impact assessment (LCIA) result' means the result of the impact category indicators in the LCIA phase (e.g., if climate change is the only impact category analysed, the LCIA result is the carbon footprint of the product or system in kg CO2 eq).

'Light means of transport battery' or 'LMT battery' means any battery that is sealed and weights below or equal to 25 kg, designed to provide electric power for the traction to wheeled vehicles that can be powered by the electric motor alone or by a combination of motor and human power including type-approved vehicle of category L in the meaning of Regulation (EU) No 168/2013, and that is not an electric vehicle battery. (Article 2 of the Battery Regulation Proposal).

'Making available on the market' means any supply of a battery for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge (Article 2 of the Battery Regulation Proposal).

'Manufacturer' means any natural or legal person who manufacturers a battery or has a battery designed or manufactured, and markets that battery under its own name or trademark or puts it into service for its own purposes (see Article 2 of the Battery Regulation Proposal).

'Most representative secondary dataset' means the dataset having the highest Technological Representativeness (TeR) according to section 6.5. If there are several datasets with the same calculated TeR, the one with the highest Geographical Representativeness (GeR) according to section 6.5 shall be selected. In the absence of information on the geographical provenience of the material, the global average shall be used.

'National accreditation body' means a national accreditation body as defined in Article 2(11) of Regulation (EC) No 765/2008 (see Article 2 of the Battery Regulation Proposal).

'Notified body' means a conformity assessment body notified in accordance with Chapter V of this Regulation (see Article 2 of the Battery Regulation Proposal).

'Partially disaggregated dataset' means a dataset with a LCI that contains elementary flows and activity data, and that only in combination with its complementing underlying datasets yield a complete aggregated LCI dataset.

'Partially disaggregated dataset at -1 level' is a partially disaggregated dataset that contains elementary flows and activity data for one level down in the supply-chain, while all complementing underlying datasets are in their aggregated form

'Placing on the market' means the first making available of a battery on the Union market (A Article 2 of the Battery Regulation Proposal).

'Rechargeable battery' means a battery that is designed to be electrically recharged (Article 2 of the Battery Regulation Proposal).

'Preparing for repurpose' means any operation, by which parts of or a complete waste battery is prepared so that it can be used for a different purpose or application than the one that it was originally designed for.

'Preparation for recycling' means treatment of waste batteries prior to any recycling process, which shall, inter alia, include storage, handling, dismantling of battery packs or separation of fractions that are not part of the battery itself (see Article 2 of the Battery Regulation Proposal).

'Producer' means any manufacturer, importer or distributor or other natural or legal person who, irrespective of the selling technique used, including by means of distance contracts as defined in Article 2(7) of

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Directive 2011/83/EU, alternatively: (i) is established in a Member State and manufactures batteries under its own name or trademark, or has batteries designed or manufactured and supplies them for the first time under its own name or trademark, including those incorporated in appliances, light means of transport or vehicles, within the territory of that Member State; (ii) is established in a Member State and resells within the territory of that Member State, under its own name or trademark, batteries, including those incorporated in appliances, light means of transport or vehicles, manufactured by others. A reseller is not regarded as the 'producer' if the brand of the manufacturer appears on the batteries, as provided for in point (i); (iii) is established in a Member State and supplies for the first time in this Member State on a professional basis, batteries, including those incorporated in appliances, light means of transport or vehicles, from a third country or from another Member State; (iv) sells batteries, including those incorporated in appliances, light means of transport or vehicles, by means of distance communication directly to end-users, that are either private households or other than private households, in a Member State, and is established in another Member State or in a third country. (Article 2 of the Battery Regulation Proposal).

'Putting into service' means the first use, for its intended purpose, in the Union, of a battery, without having been placed on the market previously (Article 2 of the Battery Regulation Proposal).

'Rechargeable battery' means a battery that is designed to be electrically recharged (Article 2 of the Battery Regulation Proposal).

'Recycler' means any natural or legal person who carries out recycling in a permitted facility (Article 2 of the Battery Regulation Proposal).

'Recycling efficiency' of a recycling process means the ratio obtained by dividing the mass of output fractions accounting for recycling by the mass of the waste batteries input fraction, expressed as a percentage (Article 2 of the Battery Regulation Proposal).

'Reference flow' means the measure of the outputs from processes in a given product system required to fulfil the function expressed by the functional unit.

'Remanufacturing' means any technical operation on a used battery that includes the disassembly and evaluation of all its battery modules and cells and the use of a certain amount of battery cells and modules, new, used or recovered from waste, or other battery components, to restore the battery capacity to at least 90% of the original rated battery capacity, and where the state of health of all individual battery cells is homogeneous, not differing more than 3% from one another, and results in the battery being used for the same purpose or application than the one for which the battery was originally designed (Article 2 of the Battery Regulation Proposal).

'Representative dataset' means a dataset having a DQR for technological representativeness (TeR) less or equal to 2 and DQR for temporal and geographical representativeness (TiR and GR) lower and equal than 3.

'Repurposing' means any operation that results in parts or the complete battery that is not a waste battery, being used for a different purpose or application than the one that the battery was originally designed for (Article 2 of the Battery Regulation Proposal).

'Residual consumption mix' is the energy source mix that is left over once the reliably tracked consumption is taken out from the generation mix. The residual consumption mix is used where an end-user is sourcing electricity from unknown origin: the energy source mix of it shall be considered as residual consumption mix. A residual consumption mix is an integral part of an energy certificate system for disclosure towards consumers, in order to prevent double counting in energy source disclosure.

'Secondary data' means data not from a specific process within the supply-chain of the company performing a carbon footprint study. This refers to data that is not directly collected, measured, or estimated by the company, but sourced from a third party LCI database or other sources. Secondary data includes industry average data (e.g., from published production data, government statistics, and industry associations), literature studies, engineering studies and patents, and may also be based on financial data, and contain proxy data, and other generic data.

'Secondary datasets' refers to any dataset that is not company-specific, e.g., from a database.

'SLI battery' means any battery designed to supply electric power for starter, lighting, or ignition and may also be used for auxiliary or backup purposes in vehicles, other means of transport or machinery; Starting, lighting and ignition (Article 2 of the Battery Regulation Proposal).

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'State of charge' means the available energy in a battery expressed as a percentage of rated capacity as declared by the manufacturer (Article 2 of the Battery Regulation Proposal).

'State of health' means a measure of the general condition of a rechargeable battery and its ability to deliver the specified performance compared with its initial condition (Article 2 of the Battery Regulation Proposal).

'Stationary battery energy storage system' means an industrial battery with internal storage specifically designed to store and deliver electric energy from and into the grid or store and deliver electric energy to end-user, regardless of where and by whom this battery is being used (Article 2 of the Battery Regulation Proposal).

'System boundary' indicates which aspects is included or excluded from the life-cycle study (Annex II of the Battery Regulation Proposal). The system boundary defines which parts of the product life-cycle and which associated life-cycle stages and processes belong to the analysed system (i.e. are required for carrying out its function as defined by the functional unit), except for those processes excluded based on the cut-off rule.

'Technical specification' means a document that prescribes technical requirements to be fulfilled by a product, process or service (Article 2 of the Battery Regulation Proposal).

'Technological proxy dataset' means process dataset describing an alternative product for which process data exist and it is assumed to have similar carbon footprint to the process in scope. Suitable proxy process datasets may differ for technological scope from the target process, but it shall involve the same life-cycle stages.

'Technologically representative secondary dataset' means that the secondary dataset has a Technological Representativeness (section 6.3.4) ≤ 4.

'Regional storage' physical place, located in the EU, where batteries are stored before they are transported to the place of installation.

'Treatment' means any activity carried out on waste batteries after they have been handed over to a facility for sorting, preparing for re-use, preparing for repurpose, preparation for recycling, or recycling' (Article 2 of the Battery Regulation Proposal).

'Tracking system' (electricity) means a system applying the process of assigning electricity generation attributes to electricity consumption.

'Unit process' means the smallest element considered in the LCI for which input and output data are quantified (based on ISO 14040:2006).

'Useable energy capacity' means the energy discharge capacity of the battery that can effectively be discharged from the battery under consideration of the limitations set by the BMS (such as maximum depth of discharge for increasing lifetime).

'Waste battery' means any battery which is waste within the meaning of Article 3(1) of Directive 2008/98/EC (Article 2 of the Battery Regulation Proposal).

The definitions of 'waste', 'waste holder', 'waste management', 'prevention', 'collection', 'separate collection', 'extended producer responsibility scheme', 'reuse', 'preparing for re-use', 'material recovery' and 'recycling' laid down in Article 3 of Directive 2008/98/EC shall apply (Article 2 of the Battery Regulation Proposal).

The definitions of 'market surveillance', 'market surveillance authority', 'fulfilment service provider', 'corrective action', 'end-user', 'recall' and 'withdrawal', as well as of 'risk' in relation to requirements of Chapters I, IV, VI, VII, IX and Annex V, Annex VIII and Annex XIII, laid down in Article 3 of Regulation (EU) 2019/1020 shall apply (Article 2 of the Battery Regulation Proposal).

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